Experimental Evaluation of a Multi-Layer Abstraction Approach for RL on Non-Markovian Tasks

Advisor: Prof. Fabio Patrizi

Master of Science in Engineering in Computer Science

Matteo Ventali

Sapienza University of Rome



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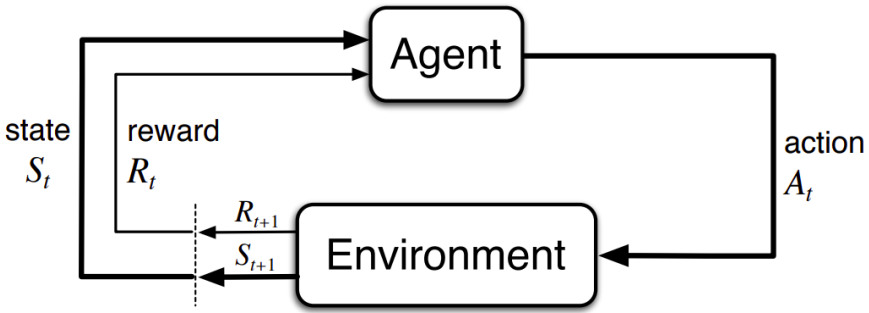


Outline

- Motivation
- Framework and Implementation
- Experimental Evaluation
- Conclusions

Motivation and Problem Statement

1 Motivation

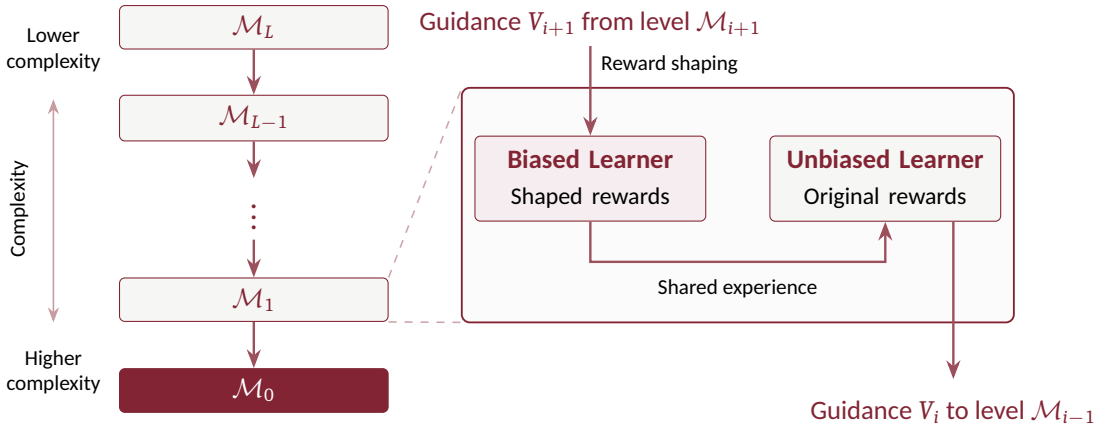


Agent–environment interaction [16].

Sparse Reward + **Non-Markovian Task** ⟶ **Prohibitive Learning**



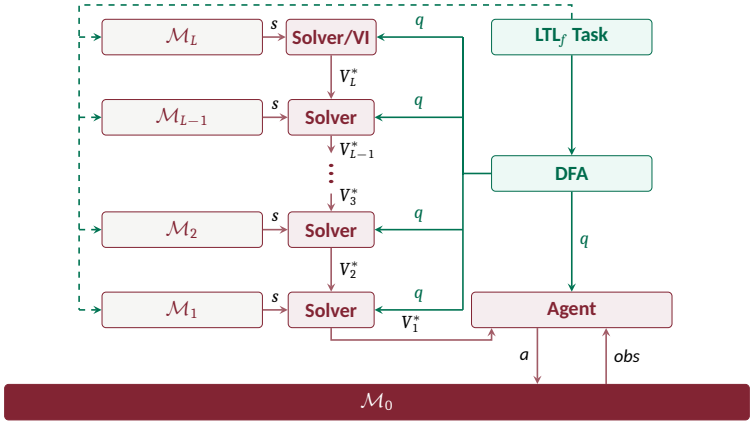
The Starting Point



Framework by Cipollone et al. [4]

Framework Proposed in This Thesis

2 Framework and Implementation



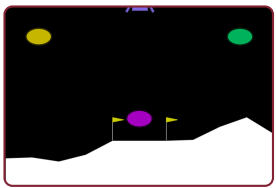
Combination of Multi-Level Framework [4] and Restraining Bolts [8].



From Concept to Implementation

2 Framework and Implementation

Task Idea



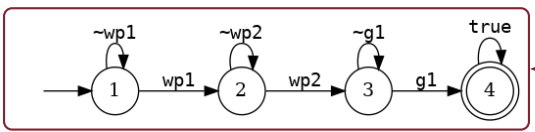
Formalization

LTL$_f$ Formula

$$\varphi = \Diamond(wp_1 \wedge \bigcirc\Diamond(wp_2 \wedge \bigcirc\Diamond g_1))$$

Translation

Equivalent DFA





Research Directions

3 Experimental Evaluation

RQ1: Direct Applicability

Can the original learning architecture be effectively applied to the considered ground domain?

RQ2: Scaling Temporal Complexity

How does abstraction-based guidance behave as temporal complexity increases?

RQ3: Learning Abstract Guidance

Can abstract guidance be learned and propagated across abstraction levels?

RQ4: Cyclic Temporal Tasks

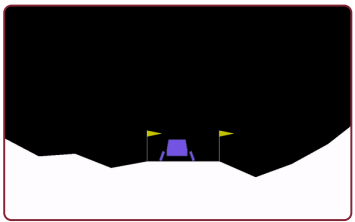
Can abstraction-based guidance support recurring temporal behaviors?

Experimental Setting

3 Experimental Evaluation

Ground Domain M_0

Gymnasium Lunar Lander

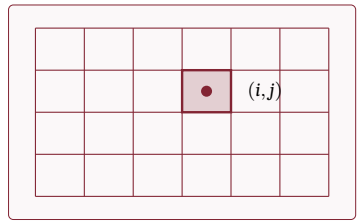


- $\mathcal{S} = (x, y, v_x, v_y, \theta, \omega, c_l, c_r)$
- $\mathcal{A} = \{\text{no-op}, \text{eng-left}, \text{eng-right}, \text{eng-main}\}$
- Original dense reward removed

State mapping

Abstract Domain M_i

Grid Environment

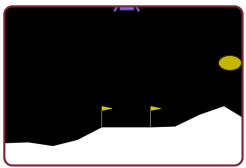


- $\mathcal{S}_i = \{(i,j) : 1 \leq i \leq H,\ 1 \leq j \leq W\}$
- $\mathcal{A} = \{\uparrow, \nearrow, \rightarrow, \searrow, \downarrow, \swarrow, \leftarrow, \nwarrow\}$
- Physical quantities discarded

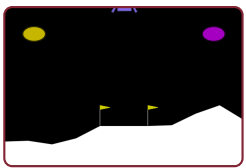
Biased–Unbiased Interaction

3 Experimental Evaluation

Episodic Tasks

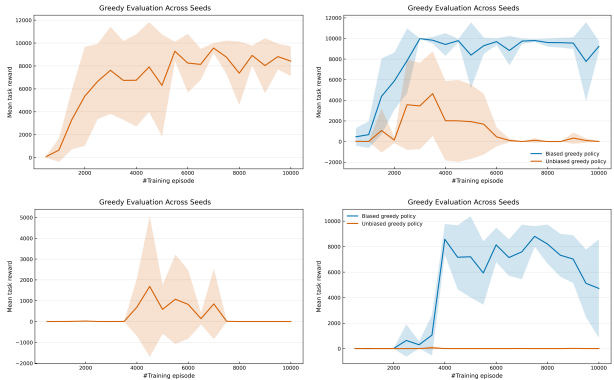


1-phase: $\Diamond g_1$



2-phase: $\Diamond(wp_1 \wedge \bigcirc\Diamond g_1)$

Training results



Greedy eval.: standalone unbiased (left), coupled biased/unbiased (right); 1-phase (top) and 2-phase (bottom).

Temporal Misalignment

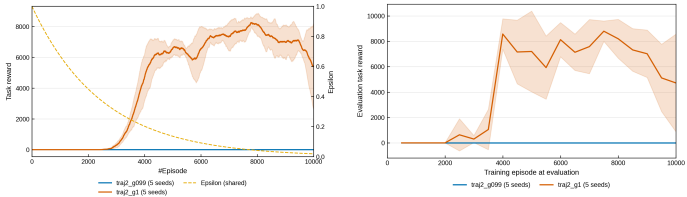
3 Experimental Evaluation

Shaping Formulation

$$\begin{cases} F(s, s') = \gamma_s \Phi(s') - \Phi(s) \\ \Phi(s) = V(s) \end{cases}$$

Potential-Based Reward Shaping (PBRS) [4, 14].

Training Results



γ_s	Best success rate (%)	Same-state fraction (%)	Same-state reward per transition	Same-state reward per episode
0.99	0.00 ± 0.00	82.39 ± 0.96	-89.51 ± 0.05	-5000.60 ± 53.21
1.00	96.94 ± 1.91	91.48 ± 0.22	0.00 ± 0.00	0.00 ± 0.00

Training reward (left) and greedy evaluation (right), with transition statistics for $\gamma_s = 0.99$ and $\gamma_s = 1$.



From Theory to Practice

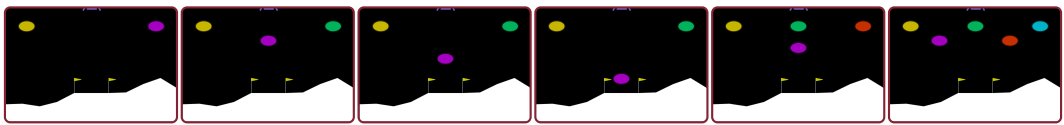
3 Experimental Evaluation

Theory		Practice
Coupled architecture Biased + Unbiased	Interaction Problem →	**Biased Learner** Only biased on the ground MDP
PBRS Formulation $F(s, s') = \gamma_s V(s') - V(s)$	Temporal Mismatch →	**Exploration Heuristic** $F(s, s') = V(s') - V(s)$

Scaling to More Complex Non-Markovian Tasks

3 Experimental Evaluation

Required Tasks



2-phase *3-phase Simple* *3-phase Hard* *3-phase Extreme* *4-phase* *5-phase*

Training Results

Task	Baseline	6×6	12×12	24×24	48×48	180×180	360×360
2-phase	16.1	91.9	97.4	96.8	98.3	95.1	97.0
3-phase Simple	5.3	95.0	94.1	98.2	98.3	98.8	96.5
3-phase Hard	0.1	95.0	93.1	97.0	88.0	98.6	97.1
3-phase Extreme	0.0	28.3	77.8	93.2	87.6	90.1	91.6
4-phase	0.0	24.8	53.5	83.6	74.9	64.2	85.1
5-phase	0.0	52.3	62.3	65.2	87.3	88.5	85.6

Best-policy success rate (%), mean over 10 training seeds.

Learning Reliability

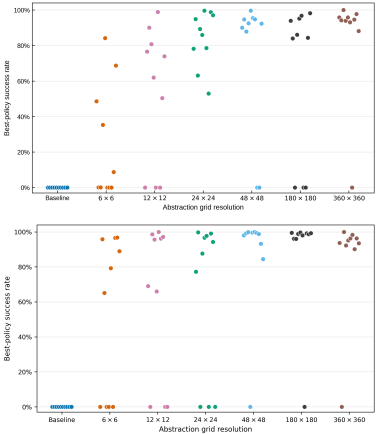
3 Experimental Evaluation

Reliability vs. Quality

Grid	4-phase N≥80%	4-phase SR ≥80%	5-phase N≥80%	5-phase SR ≥80%
6×6	1/10	84.2%	4/10	94.6%
12×12	3/10	89.9%	5/10	97.5%
24×24	6/10	94.3%	6/10	95.9%
48×48	8/10	93.4%	9/10	96.9%
180×180	7/10	91.2%	9/10	98.4%
360×360	9/10	94.8%	9/10	95.1%

N≥80%: number of high-performing seeds.
Mean SR≥80%: mean post-training best-policy success rate among high-performing seeds.

Seed-Level Distributions

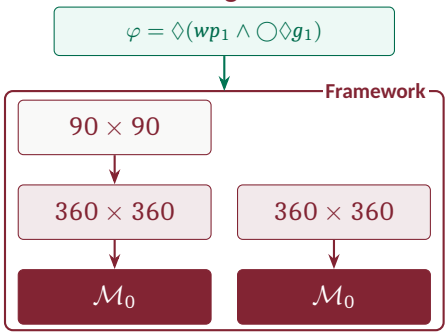


Top: 4-phase trajectory; bottom: 5-phase trajectory.
Each point represents one training seed (10 seeds per configuration).

Learning on Abstract Levels

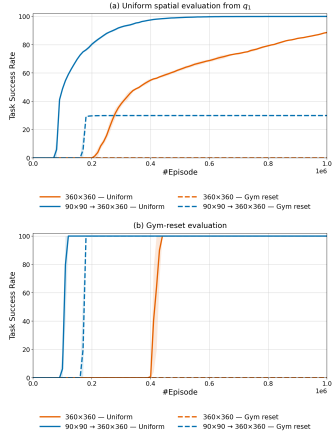
3 Experimental Evaluation

Learning Stacks



$90 \times 90 \rightarrow 360 \times 360 \rightarrow \mathcal{M}_0$ *(left) VS* $360 \times 360 \rightarrow \mathcal{M}_0$ *(right) on the 2-phase task.*

Training Results

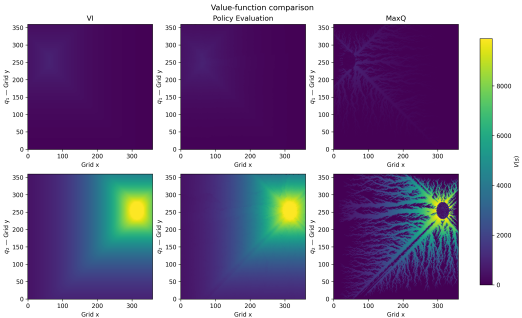


Top: uniform spatial evaluation; bottom: Gym-reset evaluation.

From a Learned Policy to an Effective Shaping Potential

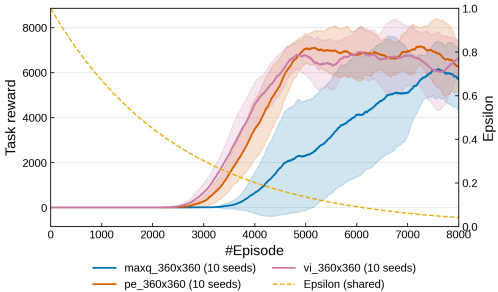
3 Experimental Evaluation

Value-Function Heatmaps



Value functions for VI, Policy Evaluation, and MaxQ (top: q_1; bottom: q_2 automaton states).

Training Results

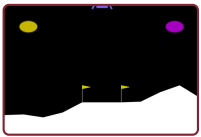


Task reward during training.

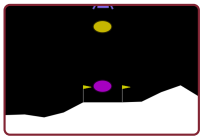
Learning on Cyclic Tasks

3 Experimental Evaluation

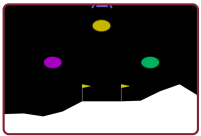
Cyclic Tasks



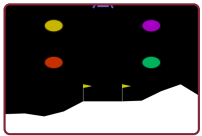
Horizontal



Vertical

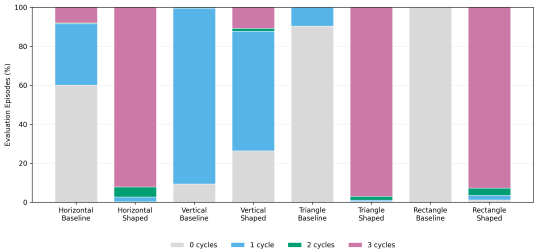


Triangle



Rectangle

Cycles Distribution

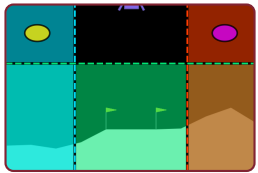


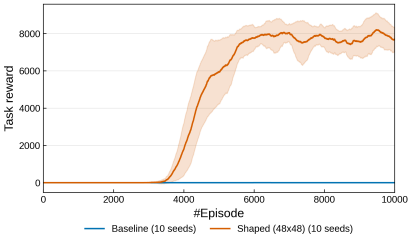
Evaluation episodes grouped by completed cycles (0–3), for baseline and shaped policies.

Additional Case Studies

3 Experimental Evaluation

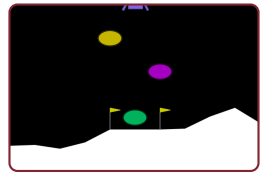
Task with Constraints

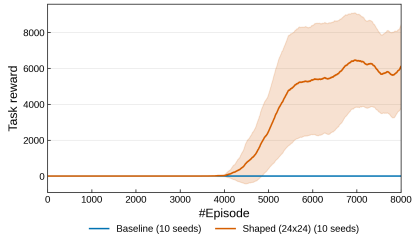




Task reward during training of 2-phase with constraints.

Landing Task





Task reward during training of landing task.



Findings, Limitations & Future Work

4 Conclusions

Findings

- Automatic guidance generation starting from logical formulas
- Effective guidance extracted from abstractions
- Multi-level architecture is promising

Limitations

- Coupled architecture interaction
- Temporal misalignment between levels
- Abstract policies could be not effective as guidance
- Long temporal tasks are difficult to be discovered

Future Work

- Study new exploration and experience-related mechanism
- Introduce different techniques across levels (e.g. planners)
- Richer logical formalism (e.g. LTL [15], STL [12])



Thank you for listening!

Any questions?

References

4 Conclusions

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